

Causality Assessment of Adverse Events

A Comprehensive Overview of WHO-UMC, Naranjo Algorithms & AEFI Principles

Executive Summary: Causality assessment evaluates the strength of association between a suspected drug/vaccine and an adverse event (AE). It is essential in pharmacovigilance for post-marketing safety signals, individual case safety report (ICSR) processing, and regulatory decision-making.

1. CORE OBJECTIVES & KEY CRITERIA

The primary aim of causality assessment is to determine whether an adverse drug reaction (ADR) was truly caused by the suspected medication or if it stems from underlying disease, drug interactions, or patient predisposition.

Temporal Relationship

The adverse event must occur within a plausible timeframe after drug administration (or vaccination).

De-challenge Outcome

Evaluates whether the clinical reaction abates or resolves upon stopping or reducing the drug dose.

Re-challenge Evidence

Observes if the adverse event recurs when the drug is re-administered under controlled conditions.

Confounding Factors

Excludes concurrent disease states, pre-existing conditions, or concomitant medications as alternative causes.

2. WHO-UMC CAUSALITY CATEGORIES

The WHO Collaborating Centre for International Drug Monitoring (Uppsala Monitoring Centre) establishes standardized clinical categories:

CATEGORY	KEY DIAGNOSTIC CRITERIA
Certain	Plausible temporal relationship; cannot be explained by disease or other drugs; positive response to withdrawal (dechallenge); pharmacologically definitive; positive rechallenge (if performed).
Probable / Likely	Reasonable temporal relationship; unlikely to be attributed to disease or other drugs; clinically reasonable response to dechallenge; rechallenge info not required.
Possible	Reasonable temporal relationship; could also be explained by concurrent disease or other drugs; information on drug withdrawal may be lacking or unclear.
Unlikely	Temporal relationship makes a causal link improbable; disease or other drugs provide plausible alternative explanations.

3. ADDITIONAL ASSESSMENT FRAMEWORKS

A. Naranjo ADR Probability Scale

A questionnaire-based scoring algorithm consisting of 10 targeted questions (scored -1, 0, +1, or +2). The cumulative score determines the causality category:

- **Definite (≥ 9 points):** High probability that the event was caused by the drug.
- **Probable (5–8 points):** Event likely drug-induced; alternative explanations improbable.
- **Possible (1–4 points):** Event could be related to drug, but alternative factors are present.
- **Doubtful (≤ 0 points):** Event is almost certainly caused by factors other than the drug.

B. WHO-AEFI Framework (Vaccine Safety)

Adverse Events Following Immunization (AEFI) utilize specialized criteria to distinguish vaccine-product related reactions, immunization error-related events, immunization anxiety-related reactions, or coincidental events.

4. ADDITIONAL PHARMACOVIGILANCE CRITERIA & FACTORS

Patient Predispositions

Underlying immunodeficiencies, autoimmune conditions, genetic polymorphism, or renal/hepatic impairment altering drug kinetics.

Pharmacological Plausibility

The reaction aligns with known target-binding pharmacology, side-effect profiles, or established class effects.

5. STANDARD CASE PROCESSING LIFECYCLE

Causality assessment is integrated into the routine Individual Case Safety Report (ICSR) processing workflow in pharmacovigilance operations:

STAGE	DESCRIPTION
1. Triage & Validation	Confirm valid 4 elements: identifiable reporter, patient, suspect drug, and adverse event.
2. Data Entry & Coding	Enter clinical history; code adverse events using MedDRA and drugs using WHO Drug Dictionary.
3. Medical Evaluation	A drug safety physician/specialist applies WHO-UMC or Naranjo tools to assign causality.
4. Regulatory Reporting	Expedited reporting to health authorities (FDA, EMA) based on seriousness and causality.

